

1. Introduction

1.1 Purpose

The main purpose of the Online Gym Management System is to automate and simplify the management of gym operations by replacing manual record-keeping with a digital system. The project aims to provide an efficient, secure, and centralized platform for handling member registration, attendance tracking, trainer management, fee collection, and report generation. Document Conventions

- Requirements are labeled as REQ-1, REQ-2, etc.
- High Priority (H), Medium Priority (M), Low Priority (L)
- Functional requirements are listed in Section 4.
- “TBD” indicates information to be finalized later.
- Bold text indicates important modules or features.

1.2 Intended Audience and Reading Suggestions

This document is intended for:

- Developers – for system design and coding
- Project Guide – for validation and approval
- Testers – for test case preparation
- Project Evaluators – for academic assessment
- Documentation Team – for report preparation

Reading sequence:

1. Introduction
2. Overall Description
3. System Features
4. Nonfunctional Requirements

1.3 Product Scope

The scope of the Online Gym Management System includes the development and implementation of a web-based platform to manage and automate the core activities of a gym or fitness center. The system covers member registration, membership plan management, attendance tracking, trainer allocation, fee collection, and report generation.

*The project is designed to serve three main users: **Admin**, **Trainer**, and **Member**.*

- *The **Admin** can add, update, or delete member and trainer records, manage membership plans, track payments, and generate financial and attendance reports.*
- *The **Trainer** can view assigned members, manage workout schedules, and monitor attendance.*
- *The **Member** can register online, view membership details, track attendance, and check payment history.*

The system ensures secure data storage using a database and provides real-time access to information. It reduces manual paperwork, improves accuracy, and enhances overall operational efficiency

2. Overall Description

2.1 Product Perspective

The **Online Gym Management System** is designed as a **standalone web-based application** that serves as a digital replacement for traditional manual gym management processes. It integrates various modules such as member management, trainer management, attendance tracking, fee collection, and reporting into a single platform, providing a unified solution for gym operations.

2.2 Product Functions

The system provides the following key functions:

1. **Member Management:** Online registration, update/delete records, track membership and renewals.
2. **Trainer Management:** Add/update trainers, assign members, manage schedules.
3. **Attendance Management:** Record daily attendance, track history, generate reports.
4. **Payment & Billing:** Record fees, generate invoices, track payment history.
5. **Reporting & Analytics:** Financial and performance reports, membership trends.
6. **User Authentication:** Secure login with role-based access for Admin, Trainer, and Member.
7. **Notifications:** Automated reminders for renewals, payments, and schedules.
8. **Search & Filter:** Quickly find members, trainers, payments, and report

2.3 User Classes and Characteristics

- **Admin:** Full access; manages members, trainers, payments, attendance, and reports.
- **Trainer:** Limited access; manages assigned members, schedules, and progress tracking.
- **Member:** Restricted access; views personal info, membership, attendance, and payment

2.4 Operating Environment

The software will operate in the following environment:

- Operating System: Windows 10/11, Linux, macOS
- Programming Language: Python 3.x
- Framework: Django
- Machine Learning Library: TensorFlow / Keras
- Image Processing: OpenCV
- Frontend: HTML, CSS
- Web Browser: Google Chrome, Microsoft Edge, Firefox
- Hardware Requirements:
 - Minimum 8GB RAM

- Intel i5 Processor or above
- GPU optional (for faster model training)

2.5 Design and Implementation Constraints

- The system must be developed using Python and CNN.
- Dataset availability may limit model accuracy.
- Project must be completed within academic timeline.
- Internet connection required for deployment.
- File upload limited to JPG and PNG formats.
- Hardware limitations may affect model training speed.

2.6 User Documentation

The following documentation will be provided:

- User Manual (PDF format)
- Installation Guide
- Admin Guide
- Project Report
- PPT Presentation
- Source Code Documentation

2.7 Assumptions and Dependencies

Assumptions:

- Users are managed.
- Dataset is properly labeled.
- Internet connection is available for deployment.

Dependencies:

- TensorFlow/Keras library
- OpenCV library
- Pre-trained CNN model
- Web hosting platform (if deployed online)

2.8 User Interfaces

The system will provide:

- **Simple web-based interface** for easy access by admin, trainers, and members
- **Login/Registration Page** for secure user authentication
- **Dashboard** displaying key details like member count, payments, and schedules
- **Membership Management Interface** to add, update, or delete member details
- **Trainer Management Section** to assign trainers and manage their schedules
- **Payment Interface** for fee submission and transaction history
- **Attendance Interface** to mark and view member attendance
- **Notification/Alert Display** for reminders about fees, sessions, or announcements

- **Error Message Display** for invalid inputs or failed operations
- **Clean and Responsive Layout** for compatibility across desktop and mobile devices
- **Navigation Menu** (Home, Members, Trainers, Payments, Attendance, Logout)

2.9 Hardware Interfaces

- Desktop/Laptop/Mobile Devices for accessing the system
- Biometric/Fingerprint Scanner (Optional) for attendance tracking
- Barcode/QR Code Scanner (Optional) for member check-in
- Server System for hosting the web application and database
- Internet Connectivity for online access and real-time update

Software Interfaces

The system interacts with:

- TensorFlow/Keras (Model processing)
- Flask (Backend server)
- OpenCV (Image preprocessing)
- NumPy (Array processing)
- Web Browser (User interaction)

2.10 Communications Interfaces

- HTTP protocol for web communication
- Localhost server or cloud hosting
- Optional REST API support
- Secure data transmission during upload

3. System Features

The system offers the following main features:

1. **Member Management**
 - Add, update, or delete member records.
 - Track membership plans and renewal dates.
2. **Trainer Management**

- Manage trainer profiles and assignments.
- Schedule workouts and monitor member progress.
- 3. **Attendance Tracking**
 - Record daily member check-ins.
 - Generate attendance reports.
- 4. **Payment & Billing**
 - Manage membership fees and additional charges.
 - Generate invoices and track payments.
- 5. **Reporting & Analytics**
 - Financial reports for revenue analysis.
 - Performance reports for members and trainers.
- 6. **User Authentication**
 - Secure login for Admin, Trainer, and Member.
 - Role-based access control.
- 7. **Notifications**
 - Automated reminders for renewals, payments, and schedules.
- 8. **Search & Filter**
 - Quick search for members, trainers, and payments.
 - Filter reports by date, trainer, or membership type

4. Nonfunctional Requirements

Nonfunctional requirements define the **quality attributes, performance, and constraints** of the system rather than specific behaviors. For the Online Gym Management System, they include:

1. **Performance**
 - The system should respond to user actions within 2–3 seconds.
 - It must handle multiple concurrent users without slowing down.
2. **Reliability**
 - The system should operate 24/7 with minimal downtime.
 - Data should be backed up regularly to prevent loss.
3. **Scalability**
 - The system should support an increasing number of members, trainers, and transactions without major modifications.
4. **Security**
 - Role-based access control for Admin, Trainer, and Member.
 - Sensitive data like personal info and payments must be encrypted.
5. **Usability**
 - Simple, intuitive, and user-friendly interface for all users.
 - Accessible via web browsers on desktops, tablets, and mobile devices.
6. **Maintainability**
 - The system should allow easy updates and bug fixes.
 - Modular design to facilitate future feature additions.
7. **Portability**
 - Compatible with common web browsers (Chrome, Firefox, Edge, Safari).
 - Minimal configuration needed for deployment on different servers

5. Project Plan

5.1 Team Members

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5.2 Division of Work

Analysis:

- Requirement gathering

Design:

- Architecture design
- Interface design

Coding:

- django Backend
- Frontend

Testing:

- Model testing
- System testing

Documentation:

- Report & PPT

5.3 Time Schedule

Week 1 – Requirement Analysis & Dataset Collection

Week 2 – Model Development & Training

Week 3 – Web Integration & Backend Development

Week 4 – Testing, Documentation & Final Deployment

Appendix A: Glossary

- **Gym Management System:** A software application used to manage gym operations such as memberships, attendance, payments, and schedules.
- **Member:** A registered user of the gym who has subscribed to a membership plan.
- **Trainer:** A professional who provides fitness training and guidance to gym members.
- **Membership Plan:** A package offered by the gym (monthly, quarterly, yearly) with specific benefits and pricing.
- **Attendance:** Record of member check-in and check-out in the gym.
- **Payment:** Transaction made by members for membership plans or services.
- **Admin:** The system administrator responsible for managing users, plans, and overall system operations.
- **Dashboard:** A user interface that displays key information such as member statistics, revenue, and schedules.
- **Schedule:** Timetable of gym activities, classes, or training sessions.
- **Database:** A structured collection of data used to store member details, payments, and records.

Appendix B: To Be Determined List

- **TBD-1: Final Number of Users**
The expected number of members and staff the system will support is yet to be finalized.
- **TBD-2: Deployment Platform**
Decision on whether the system will be deployed on a local server or cloud platform is pending.
- **TBD-3: Payment Integration Method**
Selection of payment gateway (e.g., UPI, credit/debit card) is yet to be decided.
- **TBD-4: Security Features**
Finalization of security measures such as data encryption and authentication methods is pending.
- **TBD-5: Notification System**
Whether to include SMS, email, or app notifications for reminders and updates is yet to be determined.